NW Exercise # 2.2 Questions 2.18-2.2,2.52,2.53,2.109 Exercise # 3.3 Questions 3.113- 3.135 Questions 3.62-3.78

Exercise # 2.2

For each data set in Exercises 2.18–2.23,

a. determine a frequency distribution.

b. obtain a relative-frequency distribution.

c. draw a pie chart.

d. construct a bar chart

2.18 Top Broadcast Shows

The networks for the top 20 television shows, as determined by the Nielsen Ratings for the week ending October 26, 2008, are shown below.

CBS ABC CBS ABC ABC
Fox CBS CBS Fox CBS
ABC CBS CBS CBS Fox
Fox Fox CBS Fox ABC

Tasks:

a. determine a frequency distribution.
b. obtain a relative-frequency distribution.
c. draw a pie chart.
d. construct a bar chart.

2.19 NCAA Wrestling Champs

From NCAA.com, the NCAA wrestling champions for the years 1992–2008 are:

1992 Iowa
1993 Iowa
1994 Oklahoma State
1995 Iowa
1996 Iowa
2005 Oklahoma State
2006 Oklahoma State

2007 Minnesota
2008 Iowa

Tasks:

a. determine a frequency distribution.
b. obtain a relative-frequency distribution.
c. draw a pie chart.
d. construct a bar chart.

2.20 Colleges of Students

Data on college affiliation of students in an Introduction to Computer Science course:

ENG ENG BUS BUS ENG
LIB LIB ENG ENG ENG
BUS BUS ENG BUS ENG
LIB BUS BUS BUS ENG
ENG ENG LIB ENG BUS

Tasks:

a. determine a frequency distribution.
b. obtain a relative-frequency distribution.
c. draw a pie chart.
d. construct a bar chart.

2.21 Class Levels

Class levels of students in Professor Weiss's introductory statistics course:

Jr
So So Jr Fr Jr So Jr So
So So Sr So Jr Jr Sr Fr
Jr So Jr Fr Sr Jr So
Jr Fr Fr Jr Sr
So Sr Sr So Jr So Sr So So Fr So

Tasks:

a. determine a frequency distribution.
b. obtain a relative-frequency distribution.
c. draw a pie chart.
d. construct a bar chart.

2.22 U.S. Regions

Regions of the 50 U.S. states:

SO WE WE MW NE WE WE SO MW SO
WE NE WE SO MW MW NE WE SO WE
WE SO MW SO MW WE SO NE SO SO
SO SO MW NE SO NE MW NE WE MW
WE SO MW SO MW NE MW SO NE WE

Tasks:

a. determine a frequency distribution.
b. obtain a relative-frequency distribution.
c. draw a pie chart.
d. construct a bar chart.

For each data set in Exercises 2.52–2.63, use the specified grouping method to:

a. determine a frequency distribution.
b. obtain a relative-frequency distribution.
c. construct a frequency histogram based on your result from part (a).
d. construct a relative-frequency histogram based on your result from part (b).

2.52 Number of Siblings

Professor Weiss asked his introductory statistics students to state how many siblings they have. Use single-value grouping.

1 3 2 1 1 0 1 1
3 0 2 2 1 2 0 2
1 2 2 1 0 1 1 1
1 1 0 2 0 3 4 2
0 2 1 1 2 1 1 0

Tasks:

a. determine a frequency distribution.
b. obtain a relative-frequency distribution.
c. construct a frequency histogram.
d. construct a relative-frequency histogram.

2.53 Household Size

Data on the number of people per household for a sample of 40 households. Use single-value grouping.

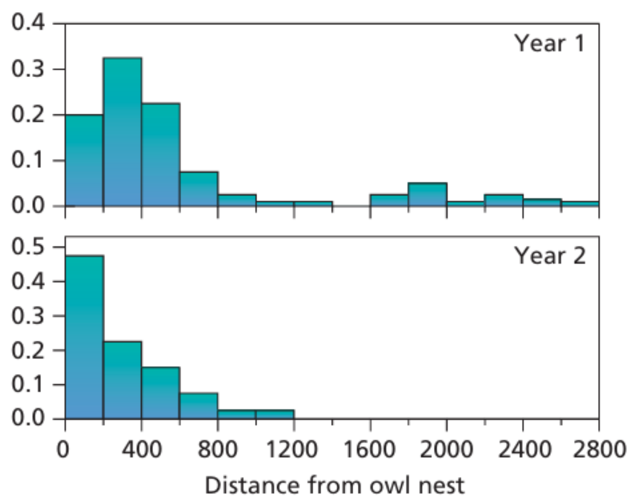
2 5 2 1 1 2 3 4
1 4 4 2 1 4 3 3
7 1 2 2 3 4 2 2
6 5 2 5 1 3 2 5
2 1 3 3 2 2 3 3

Tasks:

a. determine a frequency distribution.
b. obtain a relative-frequency distribution.
c. construct a frequency histogram.
d. construct a relative-frequency histogram.

2.109 Snow Goose Nests

The following relative-frequency histograms show the distances (in meters) of snow goose nests to the nearest snowy owl nest for two years.



Tasks for each histogram:

a. identify the overall shape of the distribution.
b. state whether the distribution is symmetric, right skewed, or left skewed.
c. compare the two distributions.

Exercise # 3.3

Exercises 3.113–3.120

We have provided simple data sets for you to practice finding the descriptive measures discussed in this section. For each data set:

a. obtain the quartiles.
b. determine the interquartile range.
c. find the five-number summary.

3.113
1, 2, 3, 4

3.114
1, 2, 3, 4, 1, 2, 3, 4

3.115
1, 2, 3, 4, 5

3.116
1, 2, 3, 4, 5, 1, 2, 3, 4, 5

3.117
1, 2, 3, 4, 5, 6

3.118
1, 2, 3, 4, 5, 6, 1, 2, 3, 4, 5, 6

3.119
1, 2, 3, 4, 5, 6, 7

3.120
1, 2, 3, 4, 5, 6, 7, 1, 2, 3, 4, 5, 6, 7

Exercises 3.121–3.128

a. obtain and interpret the quartiles.
b. determine and interpret the interquartile range.
c. find and interpret the five-number summary.
d. identify potential outliers, if any.
e. construct and interpret a boxplot.

3.121 The Great Gretzky

Wayne Gretzky, a retired professional hockey player, played 20 seasons in the National Hockey League (NHL), from 1980 through 1999. S. Berry explored some of Gretzky's accomplishments in "A Statistician Reads the Sports Pages" (*Chance*, Vol. 16, No. 1, pp. 49–54). The following table shows the number of games in which Gretzky played during each of his 20 seasons in the NHL.

79 80 80 80 74
80 80 79 64 78
73 78 74 45 81
48 80 82 82 70

3.122 Parenting Grandparents

In the article "Grandchildren Raised by Grandparents, a Troubling Trend" (*California Agriculture*, Vol. 55, No. 2, pp. 10–17), M. Blackburn considered the rates of children (under 18 years of age) living in California with grandparents as their primary caretakers. A sample of 14 California counties yielded the following percentages of children under 18 living with grandparents.

5.9 4.4 4.0 5.8 5.7 5.1 5.1

6.1 4.1 4.5 4.4 4.9 6.5 4.9

3.123 Hospital Stays

The U.S. National Center for Health Statistics compiles data on the length of stay by patients in short-term hospitals and publishes its findings in *Vital and Health Statistics*. A random sample of 21 patients yielded the following data on length of stay, in days.

4	4	12	18	9	6	12
3	6	15	7	3	55	1
10	13	5	7	1	23	9

3.124 Miles Driven

The U.S. Federal Highway Administration conducts studies on motor vehicle travel by type of vehicle. Results are published annually in *Highway Statistics*. A sample of 15 cars yields the following data on number of miles driven, in thousands, for last year.

13.2 13.3 11.9 15.7 11.3
12.2 16.7 10.7 3.3 13.6
14.8 9.6 11.6 8.7 15.0

3.125 Hurricanes

An article by D. Schaefer et al. (*Journal of Tropical Ecology*, Vol. 16, pp. 189–207) reported on a long-term study of the effects of hurricanes on tropical streams of the Luquillo Experimental Forest in Puerto Rico. The study shows that Hurricane Hugo had a significant impact on stream water chemistry. The following table shows a sample of 10 ammonia fluxes in the first year after Hugo. Data are in kilograms per hectare per year.

96 66 147 147 175
116 57 154 88 154

3.126 Sky Guide

The publication *California Wild: Natural Sciences for Thinking Animals* has a monthly feature called the "Sky Guide" that keeps track of the sunrise and sunset for the first day of each month in San Francisco. Over several issues, B. Quock from the Morrison Planetarium recorded the following sunrise times from July 1 of one year through June 1 of the next year. The times are given in minutes past midnight.

352 374 400 426 396 427
445 434 400 354 374 349

3.127 Capital Spending

An issue of *Brokerage Report* discussed the capital spending of telecommunications companies in the United States and Canada. The capital spending, in thousands of dollars, for each of 27 telecommunications companies is shown in the following table.

9,310 2,515 3,027 1,300 1,800 70 3,634 656 664 5,947 649 682 1,433 389 17,341 5,299 195 8,543
4,200 7,886 11,189 1,006 1,403 1,982 21 125 2,205

3.128 Medieval Cremation Burials

In the article "Material Culture as Memory: Combs and Cremations in Early Medieval Britain" (*Early Medieval Europe*, Vol. 12, Issue 2, pp. 89–128), H. Williams discussed the frequency of cremation burials found in 17 archaeological sites in eastern England. Here are the data.

83 64 46 48 523 35 34 265 2484
46 385 21 86 429 51 258 119

3.129 Nicotine Patches

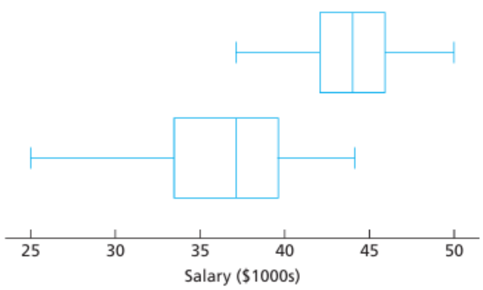
In the paper "The Smoking Cessation Efficacy of Varying Doses of Nicotine Patch Delivery Systems 4 to 5 Years Post-Quit Day" (*Preventative Medicine*, 28, pp. 113–118), D. Daughton et al. discussed the long-term effectiveness of transdermal nicotine patches on participants who had previously smoked at least 20 cigarettes per day. A sample of 15 participants in the Transdermal Nicotine Study Group (TNSG) reported that they now smoke the following number of cigarettes per day.

10 9 10 8 7
6 10 9 10 8
9 10 8 8 10

a. Determine the quartiles for these data.
b. Remark on the usefulness of quartiles with respect to this dataset.

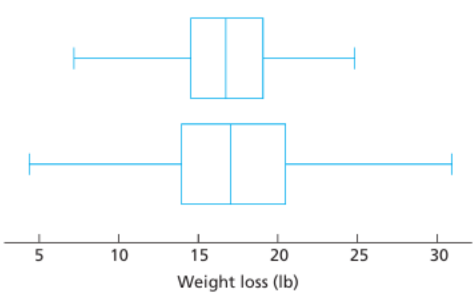
3.130 Starting Salaries

The National Association of Colleges and Employers (NACE) conducts surveys of salary offers to new college graduates and publishes the results in *Salary Survey*. The following diagram provides boxplots for the starting annual salaries, in thousands of dollars, obtained from samples of 35 business administration graduates (top boxplot) and 32 liberal arts graduates (bottom boxplot). Use the boxplots to compare the starting salaries of the sampled business administration graduates and liberal arts graduates, paying special attention to center and variation.



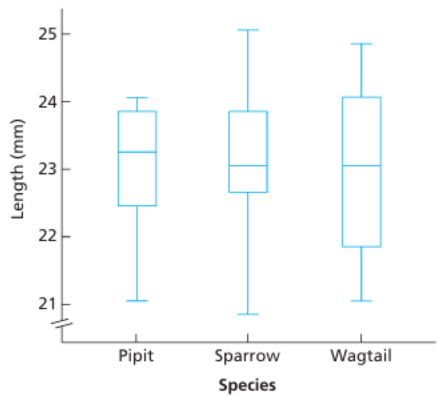
3.131 Obesity

Researchers in obesity wanted to compare the effectiveness of dieting with exercise against dieting without exercise. Seventy-three patients were randomly divided into two groups. Group 1, composed of 37 patients, was put on a program of dieting with exercise. Group 2, composed of 36 patients, dieted only. The results for weight loss, in pounds, after 2 months are summarized in the following boxplots. The top boxplot is for Group 1 and the bottom boxplot is for Group 2. Use the boxplots to compare the weight losses for the two groups, paying special attention to center and variation.



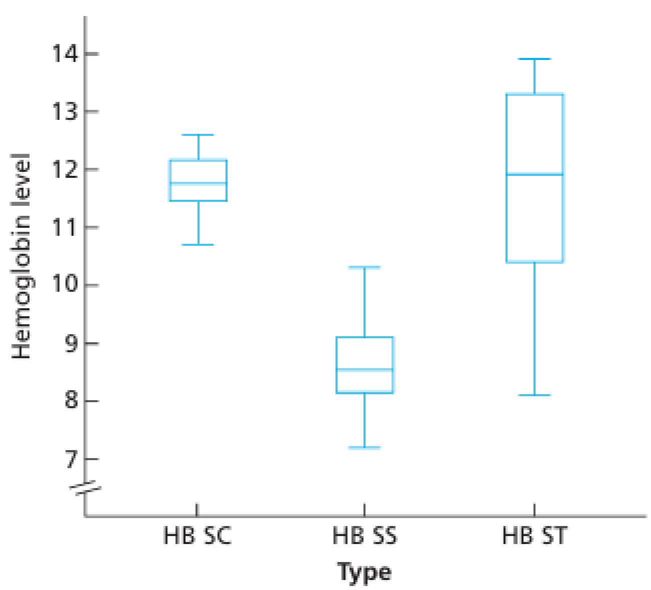
3.132 Cuckoo Care

Many species of cuckoos are brood parasites. The females lay their eggs in the nests of smaller bird species, who then raise the young cuckoos at the expense of their own young. Data on the lengths, in millimeters (mm), of cuckoo eggs found in the nests of three bird species—the Tree Pipit, Hedge Sparrow, and Pied Wagtail—were collected by the late O. M. Latter in 1902 and used by L. H. C. Tippett in his text *The Methods of Statistics* (New York: Wiley, 1952, p. 176). Use the following boxplots to compare the lengths of cuckoo eggs found in the nests of the three bird species, paying special attention to center and variation.



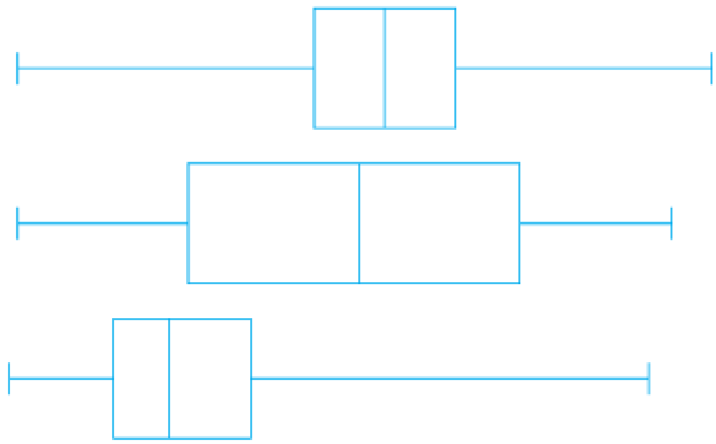
3.133 Sickle Cell Disease

A study published by E. Anionwu et al. in the *British Medical Journal* (Vol. 282, pp. 283–286) examined the steady-state hemoglobin levels of patients with three different types of sickle cell disease: HB SC, HB SS, and HB ST. Use the preceding boxplots to compare the hemoglobin levels for the three groups of patients, paying special attention to center and variation.



3.134

Each of the following boxplots was obtained from a very large data set. Use the boxplots to identify the approximate shape of the distribution of each data set.



3.135

What can you say about the boxplot of a symmetric distribution?

3.62

Consider the following four data sets.

Data Set I		Data Set II		Data Set III		Data Set IV	
1	5	1	9	5	5	2	4
1	8	1	9	5	5	4	4
2	8	1	9	5	5	4	4
2	9	1	9	5	5	4	10
5	9	1	9	5	5	4	10

a. Compute the mean of each data set.

b. Although the four data sets have the same means, in what respect are they quite different?

c. Which data set appears to have the least variation? the greatest variation?

d. Compute the range of each data set.

e. Use the defining formula to compute the sample standard deviation of each data set.

f. From your answers to parts (d) and (e), which measure of variation better distinguishes the spread in the four data sets: the range or the standard deviation? Explain your answer.

g. Are your answers from parts (c) and (e) consistent?

3.63 Age of U.S. Residents

The U.S. Census Bureau publishes information about ages of people in the United States in *Current Population Reports*. A sample of five U.S. residents have the following ages, in years.

21 54 9 45 51

a. Determine the range of these ages.

b. Find the sample standard deviation of these ages by using the defining formula, Definition 3.6 on page 105.

c. Find the sample standard deviation of these ages by using the computing formula, Formula 3.1 on page 106.

d. Compare your work in parts (b) and (c).

3.64

Consider the data set 3, 3, 3, 3, 3, 3.

a. Guess the value of the sample standard deviation without calculating it. Explain your reasoning.

b. Use the defining formula to calculate the sample standard deviation.

c. Complete the following statement and explain your reasoning:

If all observations in a data set are equal, the sample standard deviation is ________.

d. Complete the following statement and explain your reasoning:

If the sample standard deviation of a data set is 0, then ________.

Exercises 3.65–3.70

We have provided simple data sets for you to practice the basics of finding measures of variation. For each data set, determine the:

a. range.
b. sample standard deviation.

3.65
4, 0, 5

3.66
3, 5, 7

3.67
1, 2, 4, 4

3.68
2, 5, 0, −1

3.69
1, 9, 8, 4, 3

3.70
4, 2, 0, 2, 2

Exercises 3.71–3.78

Determine the range and sample standard deviation for each of the data sets. For the sample standard deviation, round each answer to one more decimal place than that used for the observations.

3.71 Amphibian Embryos

In a study of the effects of radiation on amphibian embryos titled "Shedding Light on Ultraviolet Radiation and Amphibian Embryos" (*BioScience*, Vol. 53, No. 6, pp. 551–561), L. Licht recorded the time it took for a sample of seven different species of frogs' and toads' eggs to hatch. The following table shows the times to hatch, in days.

6 7 11 6 5 5 11

3.72 Hurricanes

An article by D. Schaefer et al. (*Journal of Tropical Ecology*, Vol. 16, pp. 189–207) reported on a long-term study of the effects of hurricanes on tropical streams of the Luquillo Experimental Forest in Puerto Rico. The study showed that Hurricane Hugo had a significant impact on stream water chemistry. The following table shows a sample of 10 ammonia fluxes in the first year after Hugo. Data are in kilograms per hectare per year.

96 66 147 147 175
116 57 154 88 154

3.73 Tornado Touchdowns

Each year, tornadoes that touch down are recorded by the Storm Prediction Center and published in *Monthly Tornado Statistics*. The following table gives the number of tornadoes that touched down in the United States during each month of one year.
[SOURCE: National Oceanic and Atmospheric Administration.]

3 2 47 118 204 97
68 86 62 57 98 99

3.74 Technical Merit

In one Winter Olympics, Michelle Kwan competed in the Short Program ladies singles event. From nine judges, she received scores ranging from 1 (poor) to 6 (perfect). The following table provides the scores that the judges gave her on technical merit, found in an article by S. Berry (*Chance*, Vol. 15, No. 2, pp. 14–18).

5.8 5.7 5.9 5.7 5.5 5.7 5.7 5.7 5.6

3.75 Billionaires' Club

Each year, *Forbes* magazine compiles a list of the 400 richest Americans. As of September 17, 2008, the top 10 on the list are as shown in the following table.

Person	Wealth ($ billions)
William Gates III	57.0
Warren Buffett	50.0
Lawrence Ellison	27.0
Jim Walton	23.4
S. Robson Walton	23.3
Alice Walton	23.2
Christy Walton & family	23.2
Michael Bloomberg	20.0
Charles Koch	19.0
David Koch	19.0

3.76 AML and the Cost of Labor

Active Management of Labor (AML) was introduced in the 1960s to reduce the amount of time a woman spends in labor during the birth process. R. Rogers et al. conducted a study to determine whether AML also translates into a reduction in delivery cost to the patient. They reported their findings in the paper "Active Management of Labor: A Cost Analysis of a Randomized Controlled Trial" (*Western Journal of Medicine*, Vol. 172, pp. 240–243). The following table displays the costs, in dollars, of eight randomly sampled AML deliveries.

3141 2873 2116 1684
3470 1799 2539 3093

3.77 Fuel Economy

Every year, *Consumer Reports* publishes a magazine titled *New Car Ratings and Review* that looks at vehicle profiles for the year's models. It lets you see in one place how, within each category, the vehicles compare. One category of interest, especially when fuel prices are rising, is fuel economy, measured in miles per gallon (mpg). Following is a list of overall mpg for 14 different full-sized and compact pickups.

14 13 14 13 14 14 11
12 15 15 17 14 15 16

3.78 Router Horsepower

In the article "Router Roundup" (*Popular Mechanics*, Vol. 180, No. 12, pp. 104–109), T. Klenck reported on tests of seven fixed-base routers for performance, features, and handling. The following table gives the horsepower for each of the seven routers tested.

1.75 2.25 2.25 2.25 1.75 2.00 1.50

WP Questions 2.1-2.13 2.21-2.48.2.51,2.55,2.58 2.49-2.65 2.73-2.89 2.95-2.102

2.1

List the elements of each of the following sample spaces:

(a) the set of integers between 1 and 50 divisible by 8;

(b) the set $S = \{x \mid x^2 + 4x - 5 = 0\}$;

(c) the set of outcomes when a coin is tossed until a tail or three heads appear;

(d) the set $S = \{x \mid x \text{ is a continent}\}$;

(e) the set $S = \{x \mid 2x - 4 \geq 0 \text{ and } x < 1\}$.

2.2

Use the rule method to describe the sample space S consisting of all points in the first quadrant inside a circle of radius 3 with center at the origin.

2.3

Which of the following events are equal?

(a) $A = \{1,3\}$;

(b) $B = \{x \mid x \text{ is a number on a die}\}$;

(c) $C = \{x \mid x^2 - 4x + 3 = 0\}$;

(d) $D = \{x \mid x \text{ is the number of heads when six coins are tossed}\}$.

2.4

An experiment involves tossing a pair of dice, one green and one red, and recording the numbers that come up. If x equals the outcome on the green die and y the outcome on the red die, describe the sample space S

(a) by listing the elements (x, y);

(b) by using the rule method.

2.5

An experiment consists of tossing a die and then flipping a coin once if the number on the die is even. If the number on the die is odd, the coin is flipped twice. Using the notation 4H, for example, to denote the outcome that the die comes up 4 and then the coin comes up heads, and 3HT to denote the outcome that the die comes up 3 followed by a head and then a tail on the coin, construct a tree diagram to show the 18 elements of the sample space S.

2.6

Two jurors are selected from 4 alternates to serve at a murder trial. Using the notation A_1A_3, for example, to denote the simple event that alternates 1 and 3 are selected, list the 6 elements of the sample space S.

2.7

Four students are selected at random from a chemistry class and classified as male or female. List the elements of the sample space S_1, using the letter M for male and F for female. Define a second sample space S_2 where the elements represent the number of females selected.

2.8

For the sample space of Exercise 2.4,

(a) list the elements corresponding to the event A that the sum is greater than 8;

(b) list the elements corresponding to the event B that a 2 occurs on either die;

(c) list the elements corresponding to the event C that a number greater than 4 comes up on the green die;

(d) list the elements corresponding to the event $A \cap C$;

(e) list the elements corresponding to the event $A \cap B$;

(f) list the elements corresponding to the event $B \cap C$;

(g) construct a Venn diagram to illustrate the intersections and unions of the events A, B, and C.

2.9

For the sample space of Exercise 2.5,

(a) list the elements corresponding to the event A that a number less than 3 occurs on the die;

(b) list the elements corresponding to the event B that two tails occur;

(c) list the elements corresponding to the event A';

(d) list the elements corresponding to the event $A' \cap B$;

(e) list the elements corresponding to the event $A \cup B$.

2.10

An engineering firm is hired to determine if certain waterways in Virginia are safe for fishing. Samples are taken from three rivers.

(a) List the elements of a sample space S, using the letters F for safe to fish and N for not safe to fish.

(b) List the elements of S corresponding to event E that at least two of the rivers are safe for fishing.

(c) Define an event that has as its elements the points
$\{FFF, NFF, FFN, NFN\}$.

2.11

The resumés of two male applicants for a college teaching position in chemistry are placed in the same file as the resumés of two female applicants. Two positions become available, and the first, at the rank of assistant professor, is filled by selecting one of the four applicants at random. The second position, at the rank of instructor, is then filled by selecting at random one of the remaining three applicants. Using the notation M_2F_1, for example, to denote the simple event that the first position is filled by the second male applicant and the second position is then filled by the first female applicant,

(a) list the elements of a sample space S;

(b) list the elements of Scorresponding to event Athat the position of assistant professor is filled by a male applicant;

(c) list the elements of Scorresponding to event Bthat exactly one of the two positions is filled by a male applicant;

(d) list the elements of Scorresponding to event Cthat neither position is filled by a male applicant;

(e) list the elements of Scorresponding to the event $A \cap B$;

(f) list the elements of Scorresponding to the event $A \cup C$;

(g) construct a Venn diagram to illustrate the intersections and unions of the events A, B, and C.

2.12

Exercise and diet are being studied as possible substitutes for medication to lower blood pressure. Three groups of subjects will be used to study the effect of exercise. Group 1 is sedentary, while group 2 walks and group 3 swims for 1 hour a day. Half of each of the three exercise groups will be on a salt-free diet. An additional group of subjects will not exercise or restrict their salt, but will take the standard medication. Use Z for sedentary, W for walker, S for swimmer, Y for salt, N for no salt, M for medication, and F for medication free.

(b) Given that A is the set of nonmedicated subjects and $B$is the set of walkers, list the elements of $A \cup B$.

(c) List the elements of $A \cap B$.

2.13

Construct a Venn diagram to illustrate the possible intersections and unions for the following events relative to the sample space consisting of all automobiles made in the United States.

F: Four door
S: Sun roof
P: Power steering

2.21

Registrants at a large convention are offered 6 sightseeing tours on each of 3 days. In how many ways can a person arrange to go on a sightseeing tour planned by this convention?

2.22

In a medical study, patients are classified in 8 ways according to whether they have blood type AB+, AB−, A+, A−, B+, B−, O+, or O−, and also according to whether their blood pressure is low, normal, or high. Find the number of ways in which a patient can be classified.

2.23

If an experiment consists of throwing a die and then drawing a letter at random from the English alphabet, how many points are there in the sample space?

2.24

Students at a private liberal arts college are classified as being freshmen, sophomores, juniors, or seniors, and also according to whether they are male or female. Find the total number of possible classifications for the students of that college.

2.25

A certain brand of shoes comes in 5 different styles, with each style available in 4 distinct colors. If the store wishes to display pairs of these shoes showing all of its various styles and colors, how many different pairs will the store have on display?

2.26

A California study concluded that following 7 simple health rules can extend a man's life by 11 years on the average and a woman's life by 7 years. These 7 rules are as follows: no smoking, get regular exercise, use alcohol only in moderation, get 7 to 8 hours of sleep, maintain proper weight, eat breakfast, and do not eat between meals. In how many ways can a person adopt 5 of these rules to follow

(a) if the person presently violates all 7 rules?

(b) if the person never drinks and always eats breakfast?

2.27

A developer of a new subdivision offers a prospective home buyer a choice of 4 designs, 3 different heating systems, a garage or carport, and a patio or screened porch. How many different plans are available to this buyer?

2.28

A drug for the relief of asthma can be purchased from 5 different manufacturers in liquid, tablet, or capsule form, all of which come in regular and extra strength. How many different ways can a doctor prescribe the drug for a patient suffering from asthma?

2.29

In a fuel economy study, each of 3 race cars is tested using 5 different brands of gasoline at 7 test sites located in different regions of the country. If 2 drivers are used in the study, and test runs are made once under each distinct set of conditions, how many test runs are needed?

2.30

In how many different ways can a true-false test consisting of 9 questions be answered?

2.31

A witness to a hit-and-run accident told the police that the license number contained the letters RLH followed by 3 digits, the first of which was a 5. If the witness cannot recall the last 2 digits, but is certain that all 3 digits are different, find the maximum number of automobile registrations that the police may have to check.

2.32

(a) In how many ways can 6 people be lined up to get on a bus?

(b) If 3 specific persons, among 6, insist on following each other, how many ways are possible?

(c) If 2 specific persons, among 6, refuse to follow each other, how many ways are possible?

2.33

If a multiple-choice test consists of 5 questions, each with 4 possible answers of which only 1 is correct,

(a) in how many different ways can a student check off one answer to each question?

(b) in how many ways can a student check off one answer to each question and get all the answers wrong?

2.34

(a) How many distinct permutations can be made from the letters of the word COLUMNS?

(b) How many of these permutations start with the letter M?

2.35

A contractor wishes to build 9 houses, each different in design. In how many ways can he place these houses on a street if 6 lots are on one side of the street and 3 lots are on the opposite side?

2.36

(a) How many three-digit numbers can be formed from the digits 0, 1, 2, 3, 4, 5, and 6 if each digit can be used only once?

(b) How many of these are odd numbers?

(c) How many are greater than 330?

2.37

In how many ways can 4 boys and 5 girls sit in a row if the boys and girls must alternate?

2.38

Four married couples have bought 8 seats in the same row for a concert. In how many different ways can they be seated

(a) with no restrictions?

(b) if each couple is to sit together?

(c) if all the men sit together to the right of all the women?

2.39

In a regional spelling bee, the 8 finalists consist of 3 boys and 5 girls. Find the number of sample points in the sample space S for the number of possible orders at the conclusion of the contest for

(a) all 8 finalists;

(b) the first 3 positions.

2.40

In how many ways can 5 starting positions on a basketball team be filled with 8 men who can play any of the positions?

2.41

Find the number of ways that 6 teachers can be assigned to 4 sections of an introductory psychology course if no teacher is assigned to more than one section.

2.42

Three lottery tickets for first, second, and third prizes are drawn from a group of 40 tickets. Find the number of sample points in S for awarding the 3 prizes if each contestant holds only 1 ticket.

2.43

In how many ways can 5 different trees be planted in a circle?

2.44

In how many ways can a caravan of 8 covered wagons from Arizona be arranged in a circle?

2.45

How many distinct permutations can be made from the letters of the word INFINITY?

2.46

In how many ways can 3 oaks, 4 pines, and 2 maples be arranged along a property line if one does not distinguish among trees of the same kind?

2.47

How many ways are there to select 3 candidates from 8 equally qualified recent graduates for openings in an accounting firm?

2.48

How many ways are there that no two students will have the same birth date in a class of size 60?

2.49

Find the errors in each of the following statements:

(a) The probabilities that an automobile salesperson will sell 0, 1, 2, or 3 cars on any given day in February are, respectively, 0.19, 0.38, 0.29, and 0.15.

(b) The probability that it will rain tomorrow is 0.40, and the probability that it will not rain tomorrow is 0.52.

(c) The probabilities that a printer will make 0, 1, 2, 3, or 4 or more mistakes in setting a document are, respectively, 0.19, 0.34, −0.25, 0.43, and 0.29.

(d) On a single draw from a deck of playing cards, the probability of selecting a heart is 1/4, the probability of selecting a black card is 1/2, and the probability of selecting both a heart and a black card is 1/8.

2.50

Assuming that all elements of S in Exercise 2.8 on page 42 are equally likely to occur, find:

(a) the probability of event A;
(b) the probability of event C;
(c) the probability of event $A \cap C$.

2.51

A box contains 500 envelopes, of which 75 contain $100 in cash, 150 contain $25, and 275 contain $10. An envelope may be purchased for $25. What is the sample space for the different amounts of money? Assign probabilities to the sample points and then find the probability that the first envelope purchased contains less than $100.

2.52

Suppose that in a senior college class of 500 students it is found that 210 smoke, 258 drink alcoholic beverages, 216 eat between meals, 122 smoke and drink alcoholic beverages, 83 eat between meals and drink alcoholic beverages, 97 smoke and eat between meals, and 52 engage in all three of these bad health practices. If a member of this senior class is selected at random, find the probability that the student:

(a) smokes but does not drink alcoholic beverages;
(b) eats between meals and drinks alcoholic beverages but does not smoke;
(c) neither smokes nor eats between meals.

2.53

The probability that an American industry will locate in Shanghai, China, is 0.7, the probability that it will locate in Beijing, China, is 0.4, and the probability that it will locate in either Shanghai or Beijing or both is 0.8. What is the probability that the industry will locate:

(a) in both cities?
(b) in neither city?

2.54

From past experience, a stockbroker believes that under present economic conditions a customer will invest in tax-free bonds with a probability of 0.6, will invest in mutual funds with a probability of 0.3, and will invest in both tax-free bonds and mutual funds with a probability of 0.15. At this time, find the probability that a customer will invest:

(a) in either tax-free bonds or mutual funds;
(b) in neither tax-free bonds nor mutual funds.

2.55

If each coded item in a catalog begins with 3 distinct letters followed by 4 distinct nonzero digits, find the probability of randomly selecting one of these coded items with the first letter a vowel and the last digit even.

2.56

An automobile manufacturer is concerned about a possible recall of its best-selling four-door sedan. If there were a recall, there is a probability of 0.25 of a defect in the brake system, 0.18 of a defect in the transmission, 0.17 of a defect in the fuel system, and 0.40 of a defect in some other area.

(a) What is the probability that the defect is the brakes or the fueling system if the probability of defects in both systems simultaneously is 0.15?

(b) What is the probability that there are no defects in either the brakes or the fueling system?

2.57

If a letter is chosen at random from the English alphabet, find the probability that the letter:

(a) is a vowel exclusive of y;
(b) is listed somewhere ahead of the letter j;
(c) is listed somewhere after the letter g.

2.58

A pair of fair dice is tossed. Find the probability of getting:

(a) a total of 8;
(b) at most a total of 5.

2.59

In a poker hand consisting of 5 cards, find the probability of holding:

(a) 3 aces;
(b) 4 hearts and 1 club.

2.60

If 3 books are picked at random from a shelf containing 5 novels, 3 books of poems, and a dictionary, what is the probability that:

(a) the dictionary is selected?
(b) 2 novels and 1 book of poems are selected?

2.61

In a high school graduating class of 100 students, 54 studied mathematics, 69 studied history, and 35 studied both mathematics and history. If one of these students is selected at random, find the probability that:

(a) the student took mathematics or history;
(b) the student did not take either of these subjects;
(c) the student took history but not mathematics.

2.62

Dom's Pizza Company uses taste testing and statistical analysis of the data prior to marketing any new product. Consider a study involving three types of crusts (thin, thin with garlic and oregano, and thin with bits of cheese). Dom's is also studying three sauces (standard, a new sauce with more garlic, and a new sauce with fresh basil).

(a) How many combinations of crust and sauce are involved?

(b) What is the probability that a judge will get a plain thin crust with a standard sauce for his first taste test?

2.63

According to *Consumer Digest* (July/August 1996), the probable location of personal computers (PC) in the home is as follows:

Adult bedroom: 0.03
Child bedroom: 0.15
Other bedroom: 0.14
Office or den: 0.40
Other rooms: 0.28

(a) What is the probability that a PC is in a bedroom?
(b) What is the probability that it is not in a bedroom?
(c) Suppose a household is selected at random from households with a PC; in what room would you expect to find a PC?

2.64

Interest centers around the life of an electronic component. Suppose it is known that the probability that the component survives for more than 6000 hours is 0.42. Suppose also that the probability that the component survives no longer than 4000 hours is 0.04.

(a) What is the probability that the life of the component is less than or equal to 6000 hours?

(b) What is the probability that the life is greater than 4000 hours?

2.65

Consider the situation of Exercise 2.64. Let A be the event that the component fails a particular test and B be the event that the component displays strain but does not actually fail. Event A occurs with probability 0.20, and event B occurs with probability 0.35.

(a) What is the probability that the component does not fail the test?

(b) What is the probability that the component works perfectly well (i.e., neither displays strain nor fails the test)?

(c) What is the probability that the component either fails or shows strain in the test?

2.73

If R is the event that a convict committed armed robbery and D is the event that the convict pushed dope, state in words what probabilities are expressed by:

(a) P(R | D);
(b) P(D' | R);
(c) P(R' | D').

2.74

A class in advanced physics is composed of 10 juniors, 30 seniors, and 10 graduate students. The final grades show that 3 of the juniors, 10 of the seniors, and 5 of the graduate students received an A for the course. If a student is chosen at random from this class and is found to have earned an A, what is the probability that he or she is a senior?

2.75

A random sample of 200 adults are classified below by sex and their level of education attained.

Education	Male	Female
Elementary	38	45
Secondary	28	50
College	22	17

If a person is picked at random from this group, find the probability that:

(a) the person is a male, given that the person has a secondary education;
(b) the person does not have a college degree, given that the person is a female.

2.76

In an experiment to study the relationship of hypertension and smoking habits, the following data are collected for 180 individuals:

	Nonsmokers	Moderate Smokers	Heavy Smokers
H	21	36	30

	Nonsmokers	Moderate Smokers	Heavy Smokers
NH	48	26	19

If one of these individuals is selected at random, find the probability that the person is:

(a) experiencing hypertension, given that the person is a heavy smoker;
(b) a nonsmoker, given that the person is experiencing no hypertension.

2.77

In the senior year of a high school graduating class of 100 students, 42 studied mathematics, 68 studied psychology, 54 studied history, 22 studied both mathematics and history, 25 studied both mathematics and psychology, 7 studied history but neither mathematics nor psychology, 10 studied all three subjects, and 8 did not take any of the three. Randomly select a student from the class and find the probabilities of the following events:

(a) A person enrolled in psychology takes all three subjects.
(b) A person not taking psychology is taking both history and mathematics.

2.78

A manufacturer of a flu vaccine is concerned about the quality of its flu serum. Batches of serum are processed by three different departments having rejection rates of 0.10, 0.08, and 0.12, respectively. The inspections by the three departments are sequential and independent.

(a) What is the probability that a batch of serum survives the first departmental inspection but is rejected by the second department?
(b) What is the probability that a batch of serum is rejected by the third department?

2.79

In USA Today (Sept. 5, 1996), the results of a survey involving the use of sleepwear while traveling were listed as follows:

Type	Male	Female	Total
Underwear	0.220	0.160	0.244
Nightgown	0.002	0.180	0.182
Nothing	0.073	0.088	0.178
Pajamas	0.102	0.024	0.175

Type	Male	Female	Total
T-shirt	0.046	0.180	0.134
Other	0.084	0.018	0.087

(a) What is the probability that a traveler is a female who sleeps in the nude?
(b) What is the probability that a traveler is male?
(c) Assuming the traveler is male, what is the probability that he sleeps in pajamas?
(d) What is the probability that a traveler is male if the traveler sleeps in pajamas or a T-shirt?

2.80

The probability that an automobile being filled with gasoline also needs an oil change is 0.25; the probability that it needs a new oil filter is 0.40; and the probability that both the oil and the filter need changing is 0.14.

(a) If the oil has to be changed, what is the probability that a new oil filter is needed?
(b) If a new oil filter is needed, what is the probability that the oil has to be changed?

2.81

The probability that a married man watches a certain television show is 0.4, and the probability that a married woman watches the show is 0.5. The probability that a man watches the show, given that his wife does, is 0.7. Find the probability that:

(a) a married couple watches the show;
(b) a wife watches the show, given that her husband does;
(c) at least one member of a married couple will watch the show.

2.82

For married couples living in a certain suburb, the probability that the husband will vote on a bond referendum is 0.21, the probability that the wife will vote on the referendum is 0.28, and the probability that both the husband and the wife will vote is 0.15.

(a) What is the probability that at least one member of a married couple will vote?
(b) What is the probability that a wife will vote, given that her husband will vote?
(c) What is the probability that a husband will vote, given that his wife will not vote?

2.83

The probability that a vehicle entering the Luray Caverns has Canadian license plates is 0.12; the probability that it is a camper is 0.28; and the probability that it is a camper with Canadian license plates is 0.09. What is the probability that:

(a) a camper entering the Luray Caverns has Canadian license plates?
(b) a vehicle with Canadian license plates entering the Luray Caverns is a camper?
(c) a vehicle entering the Luray Caverns does not have Canadian plates or is not a camper?

2.84

The probability that the head of a household is home when a telemarketing representative calls is 0.4. Given that the head of the house is home, the probability that goods will be bought from the company is 0.3. Find the probability that the head of the house is home and goods are bought from the company.

2.85

The probability that a doctor correctly diagnoses a particular illness is 0.7. Given that the doctor makes an incorrect diagnosis, the probability that the patient files a lawsuit is 0.9. What is the probability that the doctor makes an incorrect diagnosis and the patient sues?

2.86

In 1970, 11% of Americans completed four years of college; 43% of them were women. In 1990, 22% of Americans completed four years of college; 53% of them were women (Time, Jan. 19, 1996).

(a) Given that a person completed four years of college in 1970, what is the probability that the person was a woman?
(b) What is the probability that a woman finished four years of college in 1990?
(c) What is the probability that a man had not finished college in 1990?

2.87

A real estate agent has 8 master keys to open several new homes. Only 1 master key will open any given house. If 40% of these homes are usually left unlocked, what is the probability that the real estate agent can get into a specific home if the agent selects 3 master keys at random before leaving the office?

2.88

Before the distribution of certain statistical software, every fourth compact disk (CD) is tested for accuracy. The testing process consists of running four independent programs and checking the results. The failure rates for the four testing programs are, respectively, 0.01, 0.03, 0.02, and 0.01.

(a) What is the probability that a CD was tested and failed any test?
(b) Given that a CD was tested, what is the probability that it failed program 2 or 3?
(c) In a sample of 100, how many CDs would you expect to be rejected?
(d) Given that a CD was defective, what is the probability that it was tested?

2.89

A town has two fire engines operating independently. The probability that a specific engine is available when needed is 0.96.

(a) What is the probability that neither is available when needed?
(b) What is the probability that a fire engine is available when needed?

2.95

In a certain region of the country it is known from past experience that the probability of selecting an adult over 40 years of age with cancer is 0.05. If the probability of a doctor correctly diagnosing a person with cancer as having the disease is 0.78 and the probability of incorrectly diagnosing a person without cancer as having the disease is 0.06, what is the probability that an adult over 40 years of age is diagnosed as having cancer?

2.96

Police plan to enforce speed limits by using radar traps at four different locations within the city limits. The radar traps at each of the locations L1, L2, L3, and L4 will be operated 40%, 30%, 20%, and 30% of the time. If a person who is speeding on her way to work has probabilities of 0.2, 0.1, 0.5, and 0.2, respectively, of passing through these locations, what is the probability that she will receive a speeding ticket?

2.97

Referring to Exercise 2.95, what is the probability that a person diagnosed as having cancer actually has the disease?

2.98

If the person in Exercise 2.96 received a speeding ticket on her way to work, what is the probability that she passed through the radar trap located at L2?

2.99

Suppose that the four inspectors at a film factory are supposed to stamp the expiration date on each package of film at the end of the assembly line. John, who stamps 20% of the packages, fails to stamp the expiration date once in every 200 packages; Tom, who stamps 60% of the packages, fails to stamp the expiration date once in every 100 packages; Jeff, who stamps 15% of the packages, fails to stamp the expiration date once in every 90 packages; and Pat, who stamps 5% of the packages, fails to stamp the expiration date once in every 200 packages. If a customer complains that her package of film does not show the expiration date, what is the probability that it was inspected by John?

2.100

A regional telephone company operates three identical relay stations at different locations. During a one-year period, the number of malfunctions reported by each station and the causes are shown below.

Cause	**Station A**	**Station B**	**Station C**
Problems with electricity supplied	2	4	1
Computer malfunction	5	3	2
Malfunctioning electrical equipment	7	4	2
Caused by other human errors	1	7	5

Suppose that a malfunction was reported and it was found to be caused by other human errors. What is the probability that it came from station C?

2.101

A paint-store chain produces and sells latex and semigloss paint. Based on long-range sales, the probability that a customer will purchase latex paint is 0.75. Of those that purchase latex paint, 60% also purchase rollers. But only 30% of semigloss paint buyers purchase rollers. A randomly selected buyer purchases a roller and a can of paint. What is the probability that the paint is latex?

2.102

Denote by A, B, and C the events that a grand prize is behind doors A, B, and C, respectively. Suppose you randomly picked a door, say A. The game host opened a door, say B, and showed there was no prize behind it. Now the host offers you the option of either staying at the door that you picked (A) or switching to the remaining unopened door (C). Use probability to explain whether you should switch or not.

2.103

A truth serum has the property that 90% of the guilty suspects are properly judged while, of course, 10% of the guilty suspects are improperly found innocent. On the other hand, innocent suspects are misjudged 1% of the time. If the suspect was selected from a group of suspects of which only 5% have ever committed a crime, and the serum indicates that he is guilty, what is the probability that he is innocent?

2.104

An allergist claims that 50% of the patients she tests are allergic to some type of weed. What is the probability that:

(a) exactly 3 of her next 4 patients are allergic to weeds?
(b) none of her next 4 patients is allergic to weeds?

2.105

By comparing appropriate regions of Venn diagrams, verify that:

(a) $(A \cap B) \cup (A \cap B') = A$

(b) $A' \cap (B' \cup C) = (A' \cap B') \cup (A' \cap C)$

MB Questions 3.1-3.24, 3.28

3.1

A computer virus is trying to corrupt two files. The first file will be corrupted with probability 0.4. Independently of it, the second file will be corrupted with probability 0.3.

(a) Compute the probability mass function (pmf) of X, the number of corrupted files.
(b) Draw a graph of its cumulative distribution function (cdf).

3.2

Every day, the number of network blackouts has a distribution (probability mass function):

x	0	1	2
P(x)	0.7	0.2	0.1

A small internet trading company estimates that each network blackout results in a $500 loss. Compute expectation and variance of this company's daily loss due to blackouts.

3.3

There is one error in one of five blocks of a program. To find the error, we test three randomly selected blocks. Let X be the number of errors in these three blocks. Compute E(X) and Var(X).

3.4

Tossing a fair die is an experiment that can result in any integer number from 1 to 6 with equal probabilities. Let X be the number of dots on the top face of a die. Compute E(X) and Var(X).

3.5

A software package consists of 12 programs, five of which must be upgraded. If 4 programs are randomly chosen for testing:

(a) What is the probability that at least two of them must be upgraded?
(b) What is the expected number of programs, out of the chosen four, that must be upgraded?

3.6

A computer program contains one error. In order to find the error, we split the program into 6 blocks and test two of them, selected at random. Let X be the number of errors in these blocks. Compute E(X).

3.7

The number of home runs scored by a certain team in one baseball game is a random variable with the distribution:

x	0	1	2
P(x)	0.4	0.4	0.2

The team plays 2 games. The number of home runs scored in one game is independent of the number of home runs in the other game. Let Y be the total number of home runs. Find E(Y) and Var(Y).

3.8

A computer user tries to recall her password. She knows it can be one of 4 possible passwords. She tries her passwords until she finds the right one. Let X be the number of wrong passwords she uses before she finds the right one. Find E(X) and Var(X).

3.9

It takes an average of 40 seconds to download a certain file, with a standard deviation of 5 seconds. The actual distribution of the download time is unknown. Using Chebyshev's inequality, what can be said about the probability of spending more than 1 minute for this download?

3.10

Every day, the number of traffic accidents has the probability mass function:

x	0	1	2	more than 2
P(x)	0.6	0.2	0.2	0

independently of other days. What is the probability that there are more accidents on Friday than on Thursday?

3.11

Two dice are tossed. Let X be the smaller number of points. Let Y be the larger number of points. If both dice show the same number, say z points, then X = Y = z.

(a) Find the joint probability mass function of (X, Y).
(b) Are X and Y independent? Explain.
(c) Find the probability mass function of X.
(d) If X = 2, what is the probability that Y = 5?

3.12

Two random variables, X and Y, have the joint distribution P(x, y):

y \ x	0	1
0	0.5	0.2
1	0.2	0.1

(a) Are X and Y independent? Explain.
(b) Are (X + Y) and (X − Y) independent? Explain.

3.13

Two random variables X and Y have the joint distribution:

P(0,0) = 0.2, P(0,2) = 0.3, P(1,1) = 0.1, P(2,0) = 0.3, P(2,2) = 0.1, and P(x,y) = 0 for all other pairs (x,y).

(a) Find the probability mass function of Z = X + Y.
(b) Find the probability mass function of U = X − Y.
(c) Find the probability mass function of V = XY.

3.14

An internet service provider charges its customers for the time of internet use rounding it up to the nearest hour. The joint distribution of the used time (X, hours) and the charge per hour (Y, cents) is given below:

y \ x	0	1	2	y \ x
0	0.52	0.20	0.04	0
1	0.14	0.02	0.01	1
2	0.06	0.01	0	2

Each customer is charged Z = X · Y cents. Find the distribution of Z.

3.15

Let X and Y be the number of hardware failures in two computer labs in a given month. The joint distribution is:

y \ x	0	1	2
0	0.52	0.20	0.04
1	0.14	0.02	0.01

y \ x	0	1	2
2	0.06	0.01	0

(a) Compute the probability of at least one hardware failure.
(b) Are X and Y independent? Why or why not?

3.16

The number of hardware failures, X, and software failures, Y, have joint distribution:

$P(0,0)=0.6$, $P(0,1)=0.1$, $P(1,0)=0.1$, $P(1,1)=0.2$.

(a) Are X and Y independent?
(b) Compute $E(X + Y)$.

3.17

Shares of company A are sold at \$10 per share. Shares of company B are sold at \$50 per share. Each share can gain \$1 with probability 0.5 or lose \$1 with probability 0.5 independently.

Which portfolio has lowest risk?

(a) 100 shares of A
(b) 50 shares of A + 10 shares of B
(c) 40 shares of A + 12 shares of B

3.18

Shares of company A cost \$10 per share and give profit X%. Shares of company B cost \$50 per share and give profit Y%.

X: $P(X = -3)=0.3$, $P(X = 0)=0.2$, $P(X = 3)=0.5$
Y: $P(Y = -3)=0.4$, $P(Y = 3)=0.6$

Compute expectations and variances of total dollar profit for:

(a) 100 shares of A
(b) 50 shares of A and 10 shares of B
(c) 20 shares of B

Find least and most risky portfolios.

3.19

A and B are two competing companies.

X (profit per share of A): $P(X=2)=0.5$, $P(X=-2)=0.5$
Y (profit per share of B): $P(Y=4)=0.2$, $P(Y=-1)=0.8$

Compute expected value and variance for:

(a) 100 shares of A
(b) 100 shares of B
(c) 50 shares of A and 50 shares of B

3.20

A quality control engineer tests computers. 5% are defective, independently.

(a) Find probability of exactly 3 defective computers in 20.
(b) Find probability that at least 5 tests are needed to find 2 defective ones.

3.21

A lab network of 20 computers is attacked by a virus. Each computer is infected with probability 0.4 independently. Find probability that at least 10 computers are infected.

3.22

5% of computer parts are defective. What is the probability that a sample of 16 contains more than 3 defective ones?

3.23

Each day, a lecture may be canceled with probability 0.05 independently.

(a) Probability at least 4 cancellations in 15 classes.
(b) Probability the 10th class is the 3rd cancellation.

3.24

A search engine checks websites; 20% contain a keyword.

(a) Probability at least 5 of first 10 contain keyword.
(b) Probability at least 5 sites are visited before first keyword appears.

3.28

Messages arrive at an electronic message center at random times, with an average of 9 messages per hour. (a) What is the probability of receiving at least five messages during the next hour? (b) What is the probability of receiving exactly five messages during the next hour?

WP Questions 5.4-5.20, 5.25-5.28, 5.56-5.58,5.67,5.71

5.4

In a certain city district, the need for money to buy drugs is stated as the reason for 75% of all thefts. Find the probability that among the next 5 theft cases reported in this district:

(a) exactly 2 resulted from the need for money to buy drugs;
(b) at most 3 resulted from the need for money to buy drugs.

5.5

The probability that a pipework failure is due to operator error is 0.30.

(a) What is the probability that out of the next 20 pipework failures at least 10 are due to operator error?
(b) What is the probability that no more than 4 out of 20 such failures are due to operator error?
(c) Suppose, for a particular plant, that out of a random sample of 20 such failures, exactly 5 are due to operator error. Do you feel that the 30% figure stated above applies to this plant? Comment.

5.6

According to a survey by the Administrative Management Society, one-half of U.S. companies give employees 4 weeks of vacation after they have been with the company for 15 years. Find the probability that among 6 companies surveyed at random, the number that give employees 4 weeks of vacation after 15 years of employment is:

(a) anywhere from 2 to 5;
(b) fewer than 3.

5.7

One prominent physician claims that 70% of those with lung cancer are chain smokers. If his assertion is correct:

(a) find the probability that of 10 such patients recently admitted to a hospital, fewer than half are chain smokers;
(b) find the probability that of 20 such patients recently admitted to a hospital, fewer than half are chain smokers.

5.8

According to a study published by a group of University of Massachusetts sociologists, approximately 60% of the Valium users in the state of Massachusetts first took Valium for psychological problems. Find the probability that among the next 8 users from this state who are interviewed:

(a) exactly 3 began taking Valium for psychological problems;
(b) at least 5 began taking Valium for problems that were not psychological.

5.9

In testing a certain kind of truck tire over rugged terrain, it is found that 25% of the trucks fail to complete the test run without a blowout. Of the next 15 trucks tested, find the probability that:

(a) from 3 to 6 have blowouts;
(b) fewer than 4 have blowouts;
(c) more than 5 have blowouts.

5.10

A nationwide survey of college seniors by the University of Michigan revealed that almost 70% disapprove of daily pot smoking. If 12 seniors are selected at random and asked their opinion, find the probability that the number who disapprove of smoking pot daily is:

(a) anywhere from 7 to 9;
(b) at most 5;
(c) not less than 8.

5.11

The probability that a patient recovers from a delicate heart operation is 0.9. What is the probability that exactly 5 of the next 7 patients having this operation survive?

5.12

A traffic control engineer reports that 75% of the vehicles passing through a checkpoint are from within the state. What is the probability that fewer than 4 of the next 9 vehicles are from out of state?

5.13

A national study that examined attitudes about antidepressants revealed that approximately 70% of respondents believe “antidepressants do not really cure anything, they just cover up the real trouble.” According to this study, what is the probability that at least 3 of the next 5 people selected at random will hold this opinion?

5.14

The percentage of wins for the Chicago Bulls basketball team going into the playoffs for the 1996–97 season was 87.7. Round the 87.7 to 90 in order to use Table A.1.

(a) What is the probability that the Bulls sweep (4–0) the initial best-of-7 playoff series?
(b) What is the probability that the Bulls win the initial best-of-7 playoff series?
(c) What very important assumption is made in answering parts (a) and (b)?

5.15

It is known that 60% of mice inoculated with a serum are protected from a certain disease. If 5 mice are inoculated, find the probability that:

(a) none contracts the disease;
(b) fewer than 2 contract the disease;
(c) more than 3 contract the disease.

5.16

Suppose that airplane engines operate independently and fail with probability equal to 0.4. Assuming that a plane makes a safe flight if at least one-half of its engines run, determine whether a 4-engine plane or a 2-engine plane has the higher probability for a successful flight.

5.17

If X represents the number of people in Exercise 5.13 who believe that antidepressants do not cure but only cover up the real problem, find the mean and variance of X when 5 people are selected at random.

5.18

(a) In Exercise 5.9, how many of the 15 trucks would you expect to have blowouts?
(b) What is the variance of the number of blowouts experienced by the 15 trucks? What does that mean?

5.19

As a student drives to school, he encounters a traffic signal. This traffic signal stays green for 35 seconds, yellow for 5 seconds, and red for 60 seconds. Assume that the student goes to school each weekday between 8:00 and 8:30 a.m. Let X1 be the number of times he encounters a green light, X2 be the number of times he encounters a yellow light, and X3 be the number of times he encounters a red light. Find the joint distribution of X1, X2, and X3.

5.20

According to USA Today (March 18, 1997), of 4 million workers in the general workforce, 5.8% tested positive for drugs. Of those testing positive, 22.5% were cocaine users and 54.4% marijuana users.

(a) What is the probability that of 10 workers testing positive, 2 are cocaine users, 5 are marijuana users, and 3 are users of other drugs?
(b) What is the probability that of 10 workers testing positive, all are marijuana users?
(c) What is the probability that of 10 workers testing positive, none is a cocaine user?

5.25

Suppose that for a very large shipment of integrated-circuit chips, the probability of failure for any one chip is 0.10. Assuming that the assumptions underlying the binomial distributions are met, find the probability that at most 3 chips fail in a random sample of 20.

5.26

Assuming that 6 in 10 automobile accidents are due mainly to a speed violation, find the probability that among 8 automobile accidents, 6 will be due mainly to a speed violation:

(a) by using the formula for the binomial distribution;
(b) by using Table A.1.

5.27

If the probability that a fluorescent light has a useful life of at least 800 hours is 0.9, find the probabilities that among 20 such lights:

(a) exactly 18 will have a useful life of at least 800 hours;
(b) at least 15 will have a useful life of at least 800 hours;
(c) at least 2 will not have a useful life of at least 800 hours.

5.28

A manufacturer knows that on average 20% of the electric toasters produced require repairs within 1 year after they are sold. When 20 toasters are randomly selected, find appropriate numbers x and y such that:

(a) the probability that at least x of them will require repairs is less than 0.5;
(b) the probability that at least y of them will not require repairs is greater than 0.8.

5.56

At a certain intersection, the number of accidents follows a Poisson model. What is the probability that in any given month at this intersection:

(a) exactly 5 accidents will occur?
(b) fewer than 3 accidents will occur?
(c) at least 2 accidents will occur?

5.57

On average, a textbook author makes two word processing errors per page on the first draft of her textbook. What is the probability that on the next page she will make:

(a) 4 or more errors?
(b) no errors?

5.58

A certain area of the eastern United States is, on average, hit by 6 hurricanes a year. Find the probability that in a given year that area will be hit by:

(a) fewer than 4 hurricanes;
(b) anywhere from 6 to 8 hurricanes.

5.67

The number of customers arriving per hour at a certain automobile service facility is assumed to follow a Poisson distribution with mean $\lambda = 7$. (a) Compute the probability that more than 10 cus tomers will arrive in a 2-hour period. (b) What is the mean number of arrivals during a 2-hour period?

5.71

For a certain type of copper wire, it is known that, on the average, 1.5 flaws occur per millimeter. Assuming that the number of flaws is a Poisson random variable, what is the probability that no flaws occur in a certain portion of wire of length 5 millimeters? What is the mean number of flaws in a portion of length 5 millimeters?

MB Questions 4.1-4.5,4.16-4.26

4.1.

The lifetime, in years, of some electronic component is a continuous random variable with the density

f(x) = { k / x^4 for x ≥ 1

{ 0 for x < 1.

Find k, the cumulative distribution function, and the probability for the lifetime to exceed 2 years.

4.2.

The time, in minutes, it takes to reboot a certain system is a continuous variable with the density

f(x) = { C(10 - x)^2, if 0 < x < 10

{ 0, otherwise

(a) Compute C.

(b) Compute the probability that it takes between 1 and 2 minutes to reboot.

4.3.

The installation time, in hours, for a certain software module has a probability density function f(x) = k(1 - x^3) for 0 < x < 1. Find k and compute the probability that it takes less than 1/2 hour to install this module.

4.4.

Lifetime of a certain hardware is a continuous random variable with density

f(x) = { K - x/50   for $0 < x < 10$ years

{ 0   for all other x

(a) Find K.

(b) What is the probability of a failure within the first 5 years?

(c) What is the expectation of the lifetime?

4.5

Two continuous random variables X and Y have the joint density

$f(x,y) = C(x^2 + y), -1 \le x \le 1, 0 \le y \le 1$

(a) Compute the constant C.
(b) Find the marginal densities of X and Y. Are these two variables independent?
(c) Compute probabilities $P\{Y < 0.6\}$ and $P\{Y < 0.6 \mid X < 0.5\}$

4.16

Let Z be a Standard Normal random variable. Compute:

(a) $P(Z < 1.25)$
(b) $P(Z \le 1.25)$
(c) $P(Z > 1.25)$
(d) $P(|Z| \le 1.25)$
(e) $P(Z < 6.0)$
(f) $P(Z > 6.0)$
(g) With probability 0.8, variable Z does not exceed what value?

4.17

For a Standard Normal random variable Z, compute:

(a) $P(Z \ge 0.99)$
(b) $P(Z \le -0.99)$
(c) $P(Z < 0.99)$
(d) $P(|Z| > 0.99)$

(e) $P(Z < 10.0)$
(f) $P(Z > 10.0)$
(g) With probability 0.9, variable Z is less than what?

4.18

For a Normal random variable X with $E(X) = -3$ and $Var(X) = 4$, compute:

(a) $P(X \leq 2.39)$
(b) $P(Z \geq -2.39)$
(c) $P(|X| \geq 2.39)$
(d) $P(|X + 3| \geq 2.39)$
(e) $P(X < 5)$
(f) $P(|X| < 5)$
(g) With probability 0.33, variable X exceeds what value?

4.19

According to one of the Western Electric rules for quality control, a produced item is considered conforming if its measurement falls within three standard deviations from the target value. Suppose that the process is in control so that the expected value of each measurement equals the target value. What percent of items will be considered conforming, if the distribution of measurements is:

(a) Normal(μ, σ)?
(b) Uniform(a, b)?

4.20

Refer to Exercise 4.19. What percent of items falls beyond 1.5 standard deviations from the mean, if the distribution of measurements is:

(a) Normal(μ, σ)?
(b) Uniform(a, b)?

4.21

The average height of professional basketball players is around 6 feet 7 inches, and the standard deviation is 3.89 inches. Assuming Normal distribution of heights within this group:

(a) What percent of professional basketball players are taller than 7 feet?
(b) If your favorite player is within the tallest 20% of all players, what can his height be?

4.22

Refer to the country in Example 4.11 on p. 91, where household incomes follow Normal distribution with $\mu = 900$ coins and $\sigma = 200$ coins:

(a) A recent economic reform made households with income below 640 coins qualify for a free bottle of milk at every breakfast. What portion of the population qualifies for a free bottle of milk?

(b) Moreover, households with an income within the lowest 5% of the population are entitled to a free sandwich. What income qualifies a household to receive free sandwiches?

4.23

The lifetime of a certain electronic component is a random variable with the expectation of 5000 hours and a standard deviation of 100 hours. What is the probability that the average lifetime of 400 components is less than 5012 hours?

4.24

Installation of some software package requires downloading 82 files. On the average, it takes 15 sec to download one file, with a variance of 16 sec^2. What is the probability that the software is installed in less than 20 minutes?

4.25

Among all the computer chips produced by a certain factory, 6 percent are defective. A sample of 400 chips is selected for inspection.

(a) What is the probability that this sample contains between 20 and 25 defective chips (including 20 and 25)?

(b) Suppose that each of 40 inspectors collects a sample of 400 chips. What is the probability that at least 8 inspectors will find between 20 and 25 defective chips in their samples?

4.26

An average scanned image occupies 0.6 megabytes of memory with a standard deviation of 0.4 megabytes. If you plan to publish 80 images on your web site, what is the probability that their total size is between 47 megabytes and 50 megabytes?

WP Questions 6.3-6.9

6.3

The daily amount of coffee, in liters, dispensed by a machine located in an airport lobby is a random variable X having a continuous uniform distribution with $A = 7$ and $B = 10$. Find the probability that on a given day the amount of coffee dispensed by this machine will be:

(a) at most 8.8 liters;
(b) more than 7.4 liters but less than 9.5 liters;
(c) at least 8.5 liters.

6.4

A bus arrives every 10 minutes at a bus stop. It is assumed that the waiting time for a particular individual is a random variable with a continuous uniform distribution.

(a) What is the probability that the individual waits more than 7 minutes?
(b) What is the probability that the individual waits between 2 and 7 minutes?

6.5

Given a standard normal distribution, find the area under the curve that lies:

(a) to the left of $z = -1.39$;
(b) to the right of $z = 1.96$;
(c) between $z = -2.16$ and $z = -0.65$;
(d) to the left of $z = 1.43$;
(e) to the right of $z = -0.89$;
(f) between $z = -0.48$ and $z = 1.74$.

6.6

Find the value of z if the area under a standard normal curve:

(a) to the right of z is 0.3622;
(b) to the left of z is 0.1131;
(c) between 0 and z, with $z > 0$, is 0.4838;
(d) between $-z$ and z, with $z > 0$, is 0.9500.

6.7

Given a standard normal distribution, find the value of k such that:

(a) $P(Z > k) = 0.2946$;
(b) $P(Z < k) = 0.0427$;
(c) $P(-0.93 < Z < k) = 0.7235$.

6.8

Given a normal distribution with $\mu = 30$ and $\sigma = 6$, find:

(a) the normal curve area to the right of $x = 17$;
(b) the normal curve area to the left of $x = 22$;
(c) the normal curve area between $x = 32$ and $x = 41$;
(d) the value of x that has 80% of the normal curve area to the left;
(e) the two values of x that contain the middle 75% of the normal curve area.

6.9

Given the normally distributed variable X with mean 18 and standard deviation 2.5, find:

(a) $P(X < 15)$;
(b) the value of k such that $P(X < k) = 0.2236$;
(c) the value of k such that $P(X > k) = 0.1814$;
(d) $P(17 < X < 21)$.

WP Questions 9.1-9.5, 9.9- 9.13

9.1

A UCLA researcher claims that the life span of mice can be extended by as much as 25% when the calories in their diet are reduced by approximately 40% from the time they are weaned. The restricted diet is enriched to normal levels by vitamins and protein. Assuming that it is known from previous studies that $\sigma = 5.8$ months, how many mice should be included in our sample if we wish to be 99% confident that the mean life span of the sample will be within 2 months of the population mean for all mice subjected to this reduced diet?

9.2

An electrical firm manufactures light bulbs that have a length of life that is approximately normally distributed with a standard deviation of 40 hours. If a sample of 30 bulbs has an average life of 780 hours, find a 96% confidence interval for the population mean of all bulbs produced by this firm.

9.3

Many cardiac patients wear an implanted pacemaker to control their heartbeat. A plastic connector module mounts on the top of the pacemaker. Assuming a standard deviation of 0.0015 inch and an approximately normal distribution, find a 95% confidence interval for the mean of the depths of all connector modules made by a certain manufacturing company. A random sample of 75 modules has an average depth of 0.310 inch.

9.4

The heights of a random sample of 50 college students showed a mean of 174.5 centimeters and a standard deviation of 6.9 centimeters.

(a) Construct a 98% confidence interval for the mean height of all college students.
(b) What can we assert with 98% confidence about the possible size of our error if we estimate the mean height of all college students to be 174.5 centimeters?

9.5

A random sample of 100 automobile owners in the state of Virginia shows that an automobile is driven on average 23,500 kilometers per year with a standard deviation of 3900 kilometers. Assume the distribution of measurements to be approximately normal.

(a) Construct a 99% confidence interval for the average number of kilometers an automobile is driven annually in Virginia.
(b) What can we assert with 99% confidence about the possible size of our error if we estimate the average number of kilometers driven by car owners in Virginia to be 23,500 kilometers per year?

9.9

Regular consumption of presweetened cereals contributes to tooth decay, heart disease, and other degenerative diseases, according to studies conducted by Dr. W. H. Bowen of the National Institute of Health and Dr. J. Yudben, Professor of Nutrition and Dietetics at the University of London. In a random sample consisting of 20 similar single servings of Alpha-Bits, the average sugar content was 11.3 grams with a standard deviation of 2.45 grams. Assuming that the sugar contents are normally distributed, construct a 95% confidence interval for the mean sugar content for single servings of Alpha-Bits.

9.10

A random sample of 12 graduates of a certain secretarial school typed an average of 79.3 words per minute with a standard deviation of 7.8 words per minute. Assuming a normal distribution for the number of words typed per minute, find a 95% confidence interval for the average number of words typed by all graduates of this school.

9.11

A machine produces metal pieces that are cylindrical in shape. A sample of pieces is taken, and the diameters are found to be 1.01, 0.97, 1.03, 1.04, 0.99, 0.98, 0.99, 1.01, and 1.03 centimeters. Find a 99% confidence interval for the mean diameter of pieces from this machine, assuming an approximately normal distribution.

9.12

A random sample of 10 chocolate energy bars of a certain brand has, on average, 230 calories per bar, with a standard deviation of 15 calories. Construct a 99% confidence interval for the true mean calorie content of this brand of energy bar. Assume that the distribution of the calorie content is approximately normal.

9.13

A random sample of 12 shearing pins is taken in a study of the Rockwell hardness of the pin head. Measurements on the Rockwell hardness are made for each of the 12, yielding an average value of 48.50 with a sample standard deviation of 1.5. Assuming the measurements to be normally distributed, construct a 90% confidence interval for the mean Rockwell hardness.

WP Questions 10.19-10.46

10.19

In a research report, Richard H. Weindruch of the UCLA Medical School claims that mice with an average life span of 32 months will live to be about 40 months old when 40% of the calories in their diet are replaced by vitamins and protein. Is there any reason to believe that $\mu < 40$ if 64 mice that are placed on this diet have an average life of 38 months with a standard deviation of 5.8 months? Use a P-value in your conclusion.

10.20

A random sample of 64 bags of white cheddar popcorn weighed, on average, 5.23 ounces with a standard deviation of 0.24 ounce. Test the hypothesis that $\mu = 5.5$ ounces against the alternative hypothesis, $\mu < 5.5$ ounces, at the 0.05 level of significance.

10.21

An electrical firm manufactures light bulbs that have a lifetime that is approximately normally distributed with a mean of 800 hours and a standard deviation of 40 hours. Test the hypothesis that $\mu = 800$ hours against the alternative, $\mu \neq 800$ hours, if a random sample of 30 bulbs has an average life of 788 hours. Use a P-value in your answer.

10.22

In the American Heart Association journal Hypertension, researchers report that individuals who practice Transcendental Meditation (TM) lower their blood pressure significantly. If a random sample of 225 male TM practitioners meditate for 8.5 hours per week with a standard deviation of 2.25 hours, does this suggest that, on average, men who use TM meditate more than 8 hours per week? Quote a P-value in your conclusion.

10.23

Test the hypothesis that the average content of containers of a particular lubricant is 10 liters if the contents of a random sample of 10 containers are 10.2, 9.7, 10.1, 10.3, 10.1, 9.8, 9.9, 10.4, 10.3, and 9.8 liters. Use a 0.01 level of significance and assume that the distribution of contents is normal.

10.24

The average height of females in the freshman class of a certain college has historically been 162.5 centimeters with a standard deviation of 6.9 centimeters. Is there reason to believe that there has been a change in the average height if a random sample of 50 females in the present freshman class has an average height of 165.2 centimeters? Use a P-value in your conclusion. Assume the standard deviation remains the same.

10.25

It is claimed that automobiles are driven on average more than 20,000 kilometers per year. To test this claim, 100 randomly selected automobile owners are asked to keep a record of the kilometers they travel. Would you agree with this claim if the random sample showed an average of 23,500 kilometers and a standard deviation of 3900 kilometers? Use a P-value in your conclusion.

10.26

According to a dietary study, high sodium intake may be related to ulcers, stomach cancer, and migraine headaches. The human requirement for salt is only 220 milligrams per day, which is surpassed in most single servings of ready-to-eat cereals. If a random sample of 20 similar servings of a certain cereal has a mean sodium content of 244 milligrams and a standard deviation of 24.5 milligrams, does this suggest at the 0.05 level of significance that the average sodium content for a single serving of such cereal is greater than 220 milligrams? Assume the distribution of sodium contents to be normal.

10.27

A study at the University of Colorado at Boulder shows that running increases the percent resting metabolic rate (RMR) in older women. The average RMR of 30 elderly women runners was 34.0% higher than the average RMR of 30 sedentary elderly women, and the standard deviations were reported to be 10.5 and 10.2%, respectively. Was there a significant increase in RMR of the women runners over the sedentary women? Assume the populations to be approximately normally distributed with equal variances. Use a P-value in your conclusions.

10.28

According to Chemical Engineering, an important property of fiber is its water absorbency. The average percent absorbency of 25 randomly selected pieces of cotton fiber was found to be 20 with a standard deviation of 1.5. A random sample of 25 pieces of acetate yielded an average percent of 12 with a standard deviation of 1.25. Is there strong evidence that the population mean percent absorbency is significantly higher for cotton fiber than for acetate? Assume that the percent absorbency is approximately normally distributed and that the population variances in percent absorbency for the two fibers are the same. Use a significance level of 0.05.

10.29

Past experience indicates that the time required for high school seniors to complete a standardized test is a normal random variable with a mean of 35 minutes. If a random sample of 20 high school seniors took an average of 33.1 minutes to complete this test with a standard deviation of 4.3 minutes, test the hypothesis, at the 0.05 level of significance, that $\mu = 35$ minutes against the alternative that $\mu < 35$ minutes.

10.30

A random sample of size $n_1 = 25$, taken from a normal population with a standard deviation $\sigma_1 = 5.2$, has a mean $\bar{x}_1 = 81$. A second random sample of size $n_2 = 36$, taken from a different normal population with a standard deviation $\sigma_2 = 3.4$, has a mean $\bar{x}_2 = 76$. Test the hypothesis that $\mu_1 = \mu_2$ against the alternative, $\mu_1 \neq \mu_2$. Quote a P-value in your conclusion.

10.31

A manufacturer claims that the average tensile strength of thread A exceeds the average tensile strength of thread B by at least 12 kilograms. To test this claim, 50 pieces of each type of thread were tested under similar conditions. Type A thread had an average tensile strength of 86.7 kilograms with a standard deviation of 6.28 kilograms, while type B thread had an average tensile strength of 77.8 kilograms with a standard deviation of 5.61 kilograms. Test the manufacturer's claim using a 0.05 level of significance.

10.32

Amstat News (December 2004) lists median salaries for associate professors of statistics at research institutions and at liberal arts and other institutions in the United States. Assume that a sample of 200 associate professors from research institutions has an average salary of $70,750 per year with a standard deviation of $6000. Assume also that a sample of 200 associate professors from other types of institutions has an average salary of $65,200 with a standard deviation of $5000. Test the hypothesis that the mean salary for associate professors in research institutions is $2000 higher than for those in other institutions. Use a 0.01 level of significance.

10.33

A study was conducted to see if increasing the substrate concentration has an appreciable effect on the velocity of a chemical reaction. With a substrate concentration of 1.5 moles per liter, the reaction was run 15 times, with an average velocity of 7.5 micromoles per 30 minutes and a standard deviation of 1.5. With a substrate concentration of 2.0 moles per liter, 12 runs were made, yielding an average velocity of 8.8 micromoles per 30 minutes and a sample standard deviation of 1.2. Is there any reason to believe that this increase in substrate concentration causes an increase in the mean velocity of the reaction of more than 0.5 micromole per 30 minutes? Use a 0.01 level of significance and assume the populations to be approximately normally distributed with equal variances.

10.34

A study was made to determine if the subject matter in a physics course is better understood when a lab constitutes part of the course. Students were randomly selected to participate in either a 3-semester-hour course without labs or a 4-semester-hour course with labs. In the section with labs, 11 students made an average grade of 85 with a standard deviation of 4.7, and in the section without labs, 17 students made an average grade of 79 with a standard deviation of 6.1. Would you say that the laboratory course increases the average grade by as much as 8 points? Use a P-value in your conclusion and assume the populations to be approximately normally distributed with equal variances.

10.35

To find out whether a new serum will arrest leukemia, 9 mice, all with an advanced stage of the disease, are selected. Five mice receive the treatment and 4 do not. Survival times, in years, from the time the experiment commenced are as follows:

Treatment 2.1 5.3 1.4 4.6 0.9
No Treatment 1.9 0.5 2.8 3.1

At the 0.05 level of significance, can the serum be said to be effective? Assume the two populations to be normally distributed with equal variances.

10.36

Engineers at a large automobile manufacturing company are trying to decide whether to purchase brand A or brand B tires for the company's new models. To help them arrive at a decision, an experiment is conducted using 12 of each brand. The tires are run until they wear out. The results are as follows:

Brand A: $\bar{x}_1$ = 37,900 kilometers, s_1 = 5100 kilometers
Brand B: $\bar{x}_2$ = 39,800 kilometers, s_2 = 5900 kilometers

Test the hypothesis that there is no difference in the average wear of the two brands of tires. Assume the populations to be approximately normally distributed with equal variances. Use a P-value.

10.37

In Exercise 9.42 on page 295, use the t distribution to test the hypothesis that the fuel economy of Volkswagen mini-trucks, on average, exceeds that of similarly equipped Toyota mini-trucks by 4 kilometers per liter. Use a 0.10 level of significance.

10.38

A UCLA researcher claims that the average life span of mice can be extended by as much as 8 months when the calories in their diet are reduced by approximately 40% from the time they are weaned. The restricted diets are enriched to normal levels by vitamins and protein. Suppose that a random sample of 10 mice is fed a normal diet and has an average life span of 32.1 months with a standard deviation of 3.2 months, while a random sample of 15 mice is fed the restricted diet and has an average life span of 37.6 months with a standard deviation of 2.8 months. Test the hypothesis, at the 0.05 level of significance, that the average life span of mice on this restricted diet is increased by 8 months against the alternative that the increase is less than 8 months. Assume the distributions of life spans for the regular and restricted diets are approximately normal with equal variances.

10.39

The following data represent the running times of films produced by two motion-picture companies:

Company 1: 102, 86, 98, 109, 92
Company 2: 81, 165, 97, 134, 92, 87, 114

Test the hypothesis that the average running time of films produced by company 2 exceeds the average running time of films produced by company 1 by 10 minutes against the one-sided alternative that the difference is less than 10 minutes. Use a 0.1 level of significance and assume the distributions of times to be approximately normal with unequal variances.

10.40

In a study conducted at Virginia Tech, the plasma ascorbic acid levels of pregnant women were compared for smokers versus nonsmokers. Thirty-two women in the last three months of pregnancy, free of major health disorders and ranging in age from 15 to 32 years, were selected for the study. Prior to the collection of 20 ml of blood, the participants were told to avoid breakfast, forego their vitamin supplements, and avoid foods high in ascorbic acid content. From the blood samples, the following plasma ascorbic acid values were determined, in milligrams per 100 milliliters:

Plasma Ascorbic Acid Values

Nonsmokers		Smokers
0.97	1.16	0.48
0.72	0.86	0.71
1.00	0.85	0.98
0.81	0.58	0.68
0.62	0.57	1.18
1.32	0.64	1.36
1.24	0.98	0.78
0.99	1.09	1.64
0.90	0.92	
0.74	0.78	
0.88	1.24	
0.94	1.18	

Is there sufficient evidence to conclude that there is a difference between plasma ascorbic acid levels of smokers and nonsmokers? Assume that the two sets of data came from normal populations with unequal variances. Use a P-value.

10.41

A study was conducted by the Department of Zoology at Virginia Tech to determine if there is a significant difference in the density of organisms at two different stations located on Cedar Run, a secondary stream in the Roanoke River drainage basin. Sewage from a sewage treatment plant and overflow from the Federal Mogul Corporation settling pond enter the stream near its headwaters. The following data give the density measurements, in number of organisms per square meter, at the two collecting stations:

Number of Organisms per Square Meter			
Station 1		Station 2	
5030	4980	2800	2810
13,700	11,910	4670	1330
10,730	8130	6890	3320
11,400	26,850	7720	1230
860	17,660	7030	2130
2200	22,800	7330	2190
4250	1130		
15,040	1690		

Can we conclude, at the 0.05 level of significance, that the average densities at the two stations are equal? Assume that the observations come from normal populations with different variances.

10.42

Five samples of a ferrous-type substance were used to determine if there is a difference between a laboratory chemical analysis and an X-ray fluorescence analysis of the iron content. Each sample was split into two subsamples and the two types of analysis were applied. Following are the coded data showing the iron content analysis:

	Sample				
Analysis	1	2	3	4	5
X-ray	2.0	2.0	2.3	2.1	2.4
Chemical	2.2	1.9	2.5	2.3	2.4

Assuming that the populations are normal, test at the 0.05 level of significance whether the two methods of analysis give, on the average, the same result.

10.43

According to published reports, practice under fatigued conditions distorts mechanisms that govern performance. An experiment was conducted using 15 college males, who were trained to make a continuous horizontal right-to-left arm movement from a microswitch to a barrier, knocking over the barrier coincident with the arrival of a clock sweep hand to the 6 o'clock position. The absolute value of the difference between the time, in milliseconds, that it took to knock over the barrier and the time for the sweep hand to reach the 6 o'clock position (500 msec) was recorded. Each participant performed the task five times under prefatigue and postfatigue conditions, and the sums of the absolute differences for the five performances were recorded.

	Absolute Time Differences	
Subject	Prefatigue	Postfatigue
1	158	91
2	92	59
3	65	215
4	98	226
5	33	223
6	89	91
7	148	92
8	58	177
9	142	134
10	117	116
11	74	153
12	66	219
13	109	143
14	57	164
15	85	100

An increase in the mean absolute time difference when the task is performed under postfatigue conditions would support the claim that practice under fatigued conditions distorts mechanisms that govern performance. Assuming the populations to be normally distributed, test this claim.

10.44

In a study conducted by the Department of Human Nutrition and Foods at Virginia Tech, the following data were recorded on sorbic acid residuals, in parts per million, in ham immediately after dipping in a sorbate solution and after 60 days of storage:

	Sorbic Acid Residuals in Ham	
Slice	Before Storage	After Storage
1	224	116
2	270	96
3	400	239
4	444	329
5	590	437
6	660	597
7	1400	689
8	680	576

Assuming the populations to be normally distributed, is there sufficient evidence, at the 0.05 level of significance, to say that the length of storage influences sorbic acid residual concentrations?

10.45

A taxi company manager is trying to decide whether the use of radial tires instead of regular belted tires improves fuel economy. Twelve cars were equipped with radial tires and driven over a prescribed test course. Without changing drivers, the same cars were then equipped with regular belted tires and driven once again over the test course. The gasoline consumption, in kilometers per liter, was recorded as follows:

	Kilometers per Liter	
Car	Radial Tires	Belted Tires
1	4.2	4.1
2	4.7	4.9
3	6.6	6.2
4	7.0	6.9
5	6.7	6.8
6	4.5	4.4
7	5.7	5.7
8	6.0	5.8
9	7.4	6.9
10	4.9	4.7
11	6.1	6.0
12	5.2	4.9

Can we conclude that cars equipped with radial tires give better fuel economy than those equipped with belted tires? Assume the populations to be normally distributed. Use a P-value in your conclusion.

10.46

In Review Exercise 9.91 on page 313, use the t distribution to test the hypothesis that the diet reduces a woman's weight by 4.5 kilograms on average against the alternative hypothesis that the mean difference in weight is less than 4.5 kilograms. Use a P-value.

WP Questions 11.1-11.9, 11.11-11.14,11.5(a, b), 11.43,11.44,11.46

Questions 11.17-11.22 Questions 13.1-13.7

11.1

A study was conducted at Virginia Tech to determine if certain static arm-strength measures have an influence on the "dynamic lift" characteristic of an individual. Twenty-five individuals were subjected to strength tests and then were asked to perform a weight lifting test in which weight was dynamically lifted over head. The data are given here.

Arm Strength, x
17.3, 19.3, 19.5, 19.7, 22.9, 23.1, 26.4, 26.8, 27.6, 28.1, 28.2, 28.7,
29.0, 29.6, 29.9, 29.9, 30.3, 31.3, 36.0, 39.5, 40.4, 44.3, 44.6, 50.4, 55.9

Dynamic Lift, y
71.7, 48.3, 88.3, 75.0, 91.7, 100.0, 73.3, 65.0, 75.0, 88.3, 68.3, 96.7,
76.7, 78.3, 60.0, 71.7, 85.0, 85.0, 88.3, 100.0, 100.0, 91.7, 100.0, 71.7

(a) Estimate β_0 and β_1 for the linear regression curve $\mu Y|x = \beta_0 + \beta_1 x$.
(b) Find a point estimate of $\mu Y|30$.
(c) Plot the residuals versus the x's (arm strength). Comment.

11.2

The grades of a class of 9 students on a midterm report (x) and on the final examination (y) are as follows:

x: 77, 50, 71, 72, 81, 94, 96, 99, 67
y: 82, 66, 78, 34, 47, 85, 99, 99, 68

(a) Estimate the linear regression line.
(b) Estimate the final examination grade of a student who received a grade of 85 on the midterm report.

11.3

The amounts of a chemical compound y that dissolved in 100 grams of water at various temperatures x were recorded as follows:

x(°C): 0, 15, 30, 45, 60, 75
y(grams):
8, 12, 25, 31, 44, 48
6, 10, 21, 33, 39, 51
8, 14, 24, 28, 42, 44

(a) Find the equation of the regression line.
(b) Graph the line on a scatter diagram.
(c) Estimate the amount of chemical that will dissolve in 100 grams of water at 50°C.

11.4

The following data were collected to determine the relationship between pressure and the corresponding scale reading for the purpose of calibration.

Pressure, x (lb/sq in.): 10, 10, 10, 10, 10, 50, 50, 50, 50, 50
Scale Reading, y: 13, 18, 16, 15, 20, 86, 90, 88, 88, 92

(a) Find the equation of the regression line.
(b) The purpose of calibration in this application is to estimate pressure from an observed scale reading. Estimate the pressure for a scale reading of 54 using
$\hat{x} = (54 - b_0)/b_1$.

11.5

A study was made on the amount of converted sugar in a certain process at various temperatures. The data were coded and recorded as follows:

Temperature, x | Converted Sugar, y
1.0 | 8.1
1.1 | 7.8
1.2 | 8.5
1.3 | 9.8
1.4 | 9.5
1.5 | 8.9
1.6 | 8.6
1.7 | 10.2
1.8 | 9.3
1.9 | 9.2
2.0 | 10.5

(a) Estimate the linear regression line.
(b) Estimate the mean amount of converted sugar produced when the coded temperature is 1.75.

11.6

In a certain type of metal test specimen, the normal stress on a specimen is known to be functionally related to the shear resistance. The following is a set of coded experimental data:

Normal Stress, x | Shear Resistance, y
26.8 | 26.5
25.4 | 27.3
28.9 | 24.2
23.6 | 27.1
27.7 | 23.6
23.9 | 25.9
24.7 | 26.3
28.1 | 22.5
26.9 | 21.7
27.4 | 21.4

22.6 | 25.8
25.6 | 24.9

(a) Estimate the regression line $\mu_{Y|x} = \beta_0 + \beta_1 x$.
(b) Estimate the shear resistance for a normal stress of 24.5.

11.7

The following is a portion of a classic dataset called the "pilot plot data" in *Fitting Equations to Data* by Daniel and Wood, published in 1971. The response y is the acid content of material produced by titration, whereas the regressor x is the organic acid content produced by extraction and weighing.

y x y x
76 37
62 82
66 88
58 43
88 109
123 48
55 138
100 164
75 28
159 70

(a) Plot the data; does it appear that a simple linear regression will be a suitable model?
(b) Fit a simple linear regression; estimate slope and intercept.
(c) Graph the regression line on the plot in (a).

11.8

A mathematics placement test is given to all entering freshmen at a small college. A student who receives a grade below 35 is denied admission to the regular mathematics course and placed in a remedial class. The placement test scores and the final grades for 20 students who took the regular course were recorded.

Placement Test | Course Grade
50 | 53
35 | 41
35 | 61
40 | 56
55 | 68
65 | 36
35 | 11
60 | 70
90 | 79

35 | 59
90 | 54
80 | 91
60 | 48
60 | 71
60 | 71
40 | 47
55 | 53
50 | 68
65 | 57
50 | 79

(a) Plot a scatter diagram.
(b) Find the equation of the regression line to predict course grades from placement test scores.
(c) Graph the line on the scatter diagram.
(d) If 60 is the minimum passing grade, below which placement test scores should students in the future be denied admission to this course?

11.9

A study was made by a retail merchant to determine the relation between weekly advertising expenditures and sales.

Advertising Costs ($) | Sales ($)
40 | 385
20 | 400
25 | 395
20 | 365
30 | 475
50 | 440
40 | 490
20 | 420
50 | 560
40 | 525
25 | 480
50 | 510

(a) Plot a scatter diagram.
(b) Find the equation of the regression line to predict weekly sales from advertising expenditures.
(c) Estimate the weekly sales when advertising costs are $35.
(d) Plot the residuals versus advertising costs. Comment.

11.11

The thrust of an engine (y) is a function of exhaust temperature (x) in °F when other important variables are held constant. Consider the following data:

y x y x
4300 1760 4010 1665
4650 1652 3810 1550
3200 1485 4500 1700
3150 1390 3008 1270
4950 1820

(a) Plot the data.
(b) Fit a simple linear regression to the data and plot the line through the data.

11.12

A study was done to study the effect of ambient temperature x on the electric power consumed by a chemical plant y. Other factors were held constant, and the data were collected from an experimental pilot plant.

y (BTU) x (°F) y (BTU) x (°F)
250 27 265 31
285 45 298 60
320 72 267 34
295 58 321 74

(a) Plot the data.
(b) Estimate the slope and intercept in a simple linear regression model.
(c) Predict power consumption for an ambient temperature of 65°F.

11.13

A study of the amount of rainfall and the quantity of air pollution removed produced the following data:

Daily Rainfall (x, 0.01 cm) Particulate Removed (y, $\mu g/m^3$)
4.3 126
4.5 121
5.9 116
5.6 118
6.1 114
5.2 118
3.8 132
2.1 141
7.5 108

(a) Find the equation of the regression line to predict the particulate removed from the amount of daily rainfall.
(b) Estimate the amount of particulate removed when the daily rainfall is x = 4.8 units.

11.14

A professor in the School of Business in a university polled a dozen colleagues about the number of professional meetings they attended in the past five years (x) and the number of papers they submitted to refereed journals (y) during the same period. The summary data are given as follows:

$n = 12$, $\bar{x} = 4$, $\bar{y} = 12$
$\sum x_i^2 = 232$
$\sum x_i y_i = 318$

Fit a simple linear regression model between x and y by finding out the estimates of intercept and slope. Comment on whether attending more professional meetings would result in publishing more papers.

11.17

Referring to Exercise 11.5 on page 398:
(a) Compute s^2.
(b) Construct a 95% confidence interval for β_0.
(c) Construct a 95% confidence interval for β_1.

11.18

Referring to Exercise 11.6 on page 399:
(a) Compute s^2.
(b) Construct a 99% confidence interval for β_0.
(c) Construct a 99% confidence interval for β_1.

11.19

Referring to Exercise 11.3 on page 398:
(a) Compute s^2.
(b) Construct a 99% confidence interval for β_0.
(c) Construct a 99% confidence interval for β_1.

11.20

Test the hypothesis $\beta_0 = 10$ using Exercise 11.8 on page 399 against the alternative $\beta_0 < 10$.
Use a 0.05 level of significance.

11.21

Test the hypothesis $\beta_1 = 6$using Exercise 11.9 on page 399 against the alternative $\beta_1 < 6$.
Use a 0.025 level of significance.

11.22

Using the value of s^2obtained in Exercise 11.16(a), construct a 95% confidence interval for $\mu_{Y|85}$in Exercise 11.2 on page 398.

11.43

Compute and interpret the correlation coefficient for the following grades of 6 students selected at random:

Mathematics grade 70 92 80 74 65 83
English grade 74 84 63 87 78 90

11.44

With reference to Exercise 11.1 on page 398, assume that x and y are random variables with a bivariate normal distribution.

(a) Calculate r.

(b) Test the hypothesis that ρ = 0 against the alternative that ρ ≠ 0 at the 0.05 level of significance.

11.46

Test the hypothesis that ρ = 0 in Exercise 11.43 against the alternative that ρ ≠ 0. Use a 0.05 level of significance.

13.1

Six different machines are being considered for use in manufacturing rubber seals. The machines are compared based on the tensile strength of the product. A random sample of four seals from each machine is taken to determine whether the mean tensile strength differs among machines. The tensile strength measurements ($kg/cm^2 \times 10^{-1}$) are given below:

Machine	1	2	3	4	5	6
Values	17.5, 16.9, 15.8, 18.6	16.4, 19.2, 17.7, 15.4	20.3, 15.7, 17.8, 18.9	14.6, 16.7, 20.8, 18.9	17.5, 19.2, 16.5, 20.5	18.3, 16.2, 17.5, 20.1

Perform a one-way ANOVA at the 0.05 level of significance and determine whether there is a significant difference in mean tensile strengths among the six machines.

13.2

The data below represent the number of hours of relief provided by five different brands of headache tablets administered to 25 subjects with fever of 38°C or higher. Perform a one-way ANOVA at the 0.05 level of significance to test whether the mean duration of relief is the same for all five brands. Discuss the results.

Tablet A	Tablet B	Tablet C	Tablet D	Tablet E
5.2	9.1	3.2	2.4	7.1
4.7	7.1	5.8	3.4	6.6
8.1	8.2	2.2	4.1	9.3
6.2	6.0	3.1	1.0	4.2
3.0	9.1	7.2	4.0	7.6

13.3

A study titled *"Shelf-Space Strategy in Retailing"* investigates the effect of shelf height on supermarket sales of canned dog food. An experiment was conducted in a small supermarket over 8 days. Each day, shelf height was randomly changed three times among knee level, waist level, and eye level.

Sales (in hundreds of dollars) of Arf dog food for each shelf height are shown below:

Knee Level	Waist Level	Eye Level
77	88	85

Knee Level	Waist Level	Eye Level
82	94	85
86	93	87
78	90	81
81	91	80
86	94	79
77	90	87
81	87	93

Using a 0.01 level of significance, determine whether there is a significant difference in mean daily sales among shelf heights.

13.4

An experiment was conducted to study the knockdown time (time from injection to immobilization) of three different immobilizing drugs used on wild white-tailed deer. Thirty male deer were randomly assigned to three treatment groups:

- Group A: 5 mg liquid succinylcholine chloride (SCC)
- Group B: 8 mg powdered SCC
- Group C: 200 mg phencyclidine hydrochloride

Knockdown times (in minutes) are shown below:

Group A	Group B	Group C
11	11	23
5	10	4
14	7	11
7	10	6
10	4	12
7	6	4

Group A	Group B	Group C
16	7	3
7	10	5
6	8	3
12	7	7

Perform a one-way ANOVA at the 0.01 level of significance to determine whether the mean knockdown times differ among the three drugs.

13.5

A study on the enzyme NADPH:NAD transhydrogenase of the tapeworm *Hymenolepiasis diminuta* examines changes in enzyme activity under different concentrations of NADP. Enzyme specific activity (nmol/min/mg protein) was measured under four concentrations:

NADP Concentration (nm)	0	80	160	360
Values	11.01, 12.09, 10.55, 11.26, 9.36	11.38, 10.67, 12.33, 10.08	11.02, 10.67, 11.50, 10.31	6.04, 8.65, 7.76, 10.13

Test at the 0.01 level of significance whether the mean enzyme activity is the same across all four concentrations.

13.6

A study measures sorption rates of three types of organic solvents used in industrial cleaning processes. Independent samples were taken and measured as mole percentage.

Aromatics	Chloroalkanes	Esters
1.06, 0.79, 0.82, 0.89, 1.05	0.95, 0.65, 1.15, 1.12, 0.61	1.58, 1.45, 0.57, 1.16, 0.34
1.12, 0.91, 0.83, 0.43, 0.60	0.29, 0.06, 0.44, 0.55	0.06, 0.09, 0.17, 0.17

Determine whether there is a significant difference in mean sorption rates using a P-value approach. Which solvent would be preferred?

13.7

A study investigates the effect of magnesium ammonium phosphate ($MgNH_4PO_4$) on the height of chrysanthemum plants. Forty seedlings were divided into four groups of ten plants each and treated with different concentrations (g/bu):

50 g/bu	**100 g/bu**	**200 g/bu**	**400 g/bu**
13.2, 12.8, 13.0, 14.2, 15.0	12.4, 17.2, 14.0, 21.6, 20.0	16.0, 14.8, 14.0, 14.0, 22.2	12.6, 13.0, 23.6, 17.0, 24.4
7.8, 20.0, 17.0, 19.6, 20.2	14.4, 15.8, 27.0, 18.0, 23.2	21.0, 19.1, 18.0, 21.1, 25.0	14.8, 15.8, 26.0, 22.0, 18.2

At the 0.05 level of significance, determine whether different concentrations affect mean plant height. Identify the most effective concentration.

NW Questions 16.42-16.47

Exercises 16.42–16.47

In Exercises 16.42–16.47, data are provided from independent simple random samples from several populations. In each case:

a. Compute SST, SSTR, and SSE using the computational formulas given in Formula 16.1 on page 726.
b. Compare your results in part (a) for SSTR and SSE with those obtained in Exercises 16.24–16.29, where the defining formulas were used.
c. Construct a one-way ANOVA table.
d. At the 5% level of significance, determine whether there is sufficient evidence to conclude that not all population means are equal.

16.42

Sample 1	Sample 2	Sample 3
1	10	4
9	4	16
	8	10
	6	
	2	

16.43

Sample 1	Sample 2	Sample 3
8	2	4
4	1	3
6	3	6
		3

16.44

Sample 1	Sample 2	Sample 3	Sample 4
6	9	4	8
3	5	4	4
3	7	2	6
	8	2	
	6	3	

16.45

Sample 1	Sample 2	Sample 3	Sample 4	Sample 5
7	5	6	3	7
4	9	7	7	9
5	4	5	7	11
4		4	4	
		8	4	

16.46

Sample 1	Sample 2	Sample 3	Sample 4	Sample 5
4	8	9	4	3
2	5	6	0	6
3	5	9	2	9

16.47

Sample 1	Sample 2	Sample 3	Sample 4
11	9	16	5
6	2	10	1
7	4	10	3